

Simultaneous Rotational Alignment of Multi-Core Fibre Ribbons for Scalable Interconnection

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Abstract We propose a simultaneous rotational alignment method for multi-core fibre ribbons, enabling scalable interconnection with reduced assembly time. Multiple fibres are aligned and selectively fixed during collective rotation using a dedicated jig. We experimentally demonstrate an average rotational misalignment of 0.83°. ©2026 The Author(s)

Introduction

Multi-core fibre (MCF) has attracted significant attention as a promising technology for further scaling the transmission capacity of optical communication systems [1]. In addition to transmission demonstrations, recent progress in standardisation activities by international organisations such as the International Electrotechnical Commission (IEC) and International Telecommunication Union Telecommunication Standardization Sector (ITU-T) has accelerated the transition of MCF from laboratory research to practical deployment [2-4]. As a result, manufacturing and assembly technologies for MCF-based components are becoming increasingly important for real-world applications.

Figure 1 illustrates an overview of MCF Interconnection. Unlike conventional single-mode fibre (SMF), MCF incorporates multiple spatially separated cores within a single cladding, requiring precise rotational alignment during interconnection to ensure low-loss connection across all cores [5-6]. This alignment process, often referred to as core alignment, becomes particularly challenging when connecting multiple MCFs, such as in MCF ribbons [7-9]. In conventional approaches, each MCF is aligned, fixed, and bonded individually, resulting in an assembly time that increases linearly with the number of fibres. Although this serial procedure is straightforward, it leads to substantial process overhead and limits scalability for high-core-count or mass-production scenarios. However, a practical method to align multiple MCFs simultaneously while maintaining rotational accuracy has not yet been established.

In this paper, we propose a simultaneous rotational alignment method for MCF ribbons, in which all fibres are rotated collectively while only the fibres that have reached their optimum alignment are selectively fixed. By combining a rotatable holder with independent temporary fixation on V-grooves, the proposed method enables scalable interconnection with significantly reduced assembly time. Experimental results using 4-core MCF ribbons demonstrate the feasibility of

the proposed approach.

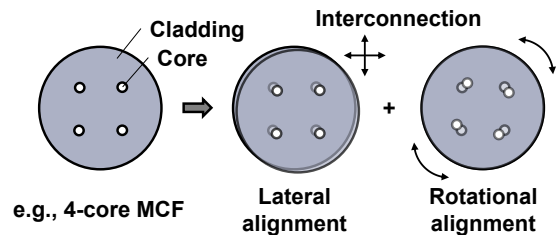


Fig. 1: Overview of MCF Interconnection.

Comparison of Conventional and Simultaneous Rotational Alignment Method

Figure 2 illustrates a comparison between the conventional serial alignment approach and the proposed simultaneous rotational alignment method for MCF ribbons.

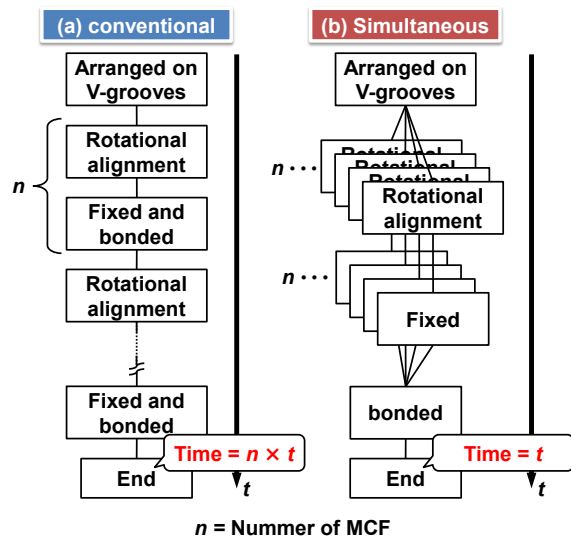


Fig. 2: Conventional vs. Simultaneous rotational alignment. Conventional rotational alignment method (a), and Simultaneous rotational alignment method (b).

In the conventional approach (Fig. 2(a)), all MCFs are first arranged on V-grooves to define their lateral positions. Subsequently, rotational alignment is performed for one MCF at a time. The selected MCF is rotated to achieve optimum core alignment, then fixed and bonded before proceeding to the alignment of the next MCF.

This procedure is repeated sequentially for all MCFs in the ribbon, forming a strictly serial alignment process. As a result, the total assembly time increases linearly with the number of fibres, which limits scalability for high-fibre-count MCF ribbon assembly.

In contrast, the proposed method (Fig. 2(b)) performs rotational alignment of multiple MCFs simultaneously. All MCFs undergo collective rotation, while the rotational alignment condition of each MCF is evaluated independently. MCFs that reach their optimum alignment are selectively fixed without interrupting the rotation of the remaining MCFs. Through this selective fixation during a single rotational scan, all MCFs in the ribbon can be aligned and fixed efficiently. This fundamental difference in alignment flow enables the proposed method to significantly reduce assembly time in multi-fibre connector fabrication using MCFs and MCF splicing processes, while maintaining the rotational accuracy required for low-loss connection.

Overview of the 4-core MCF ribbon

Among various space-division multiplexing (SDM) fibres, standard outer-diameter ($\approx 125\ \mu\text{m}$) MCF has been actively discussed in international standardization activities at the IEC and ITU-T and is regarded as one of the most practical candidates for deployment [10–11]. Accordingly, a 12-fibre ribbon based on standard outer-diameter 4-core MCFs is employed in this study.

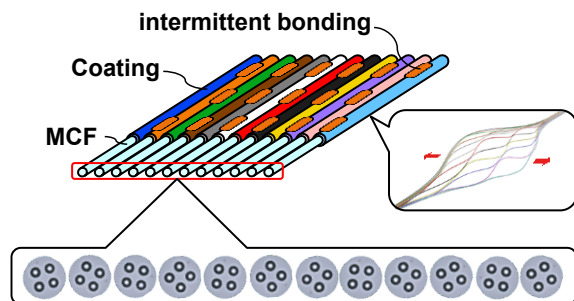


Fig. 3: Configuration of a multi-core 12-fibre intermittent ribbon composed of 4-core MCFs with a standard cladding outer diameter of $125\ \mu\text{m}$.

Figure 3 illustrates an overview of the 12-MCF ribbon used in this study. The ribbon is configured as an intermittent ribbon structure, in which individual MCFs are periodically bonded to form a ribbon while maintaining flexible fibre sections between bonded segments. This intermittent ribbon configuration provides sufficient positional stability for handling and alignment, while preserving mechanical flexibility of the fibres. Such a structure is suitable for experimental evaluation of the proposed alignment method, as it allows multiple MCFs to be arranged in a ribbon form without constraining their rotational degrees of freedom.

Experimental Setup for Simultaneous Rotational Alignment Method

Figure 4 illustrates the experimental setup used to evaluate the proposed simultaneous rotational alignment method. The setup is built around a custom-fabricated alignment jig designed to enable collective rotation of multiple MCFs while allowing independent fixation of each fibre. As shown in Fig. 4, the MCF ribbon is placed on a set of V-grooves that define the lateral positions of individual MCFs. The V-grooves provide stable positioning without constraining the rotational degrees of freedom of the fibres. All MCFs are held by a common holder, enabling collective rotational motion during the alignment process. The jig is configured such that rotational alignment and fibre fixation are mechanically decoupled. While all MCFs are subjected to the same rotational motion, the fixation of each MCF can be performed independently. This configuration allows an MCF that has reached its optimum rotational alignment to be fixed without interrupting the rotation of the remaining unfixed MCFs. By this structural design, the experimental setup realizes the alignment flow described in Fig. 2(b), in which multiple MCFs are aligned simultaneously through collective rotation, followed by selective fixation. The detailed fixation mechanism and the evaluation procedure of rotational alignment are described in the following subsections.

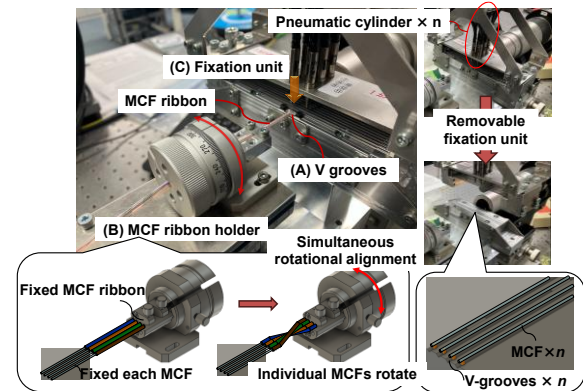


Fig. 4: Overview of MCF ribbon alignment jig.

Figure 5 provides a detailed view of the fibre fixation mechanism employed in the proposed alignment jig. The mechanism is designed to independently fix each MCF after it reaches its optimum rotational alignment, while preserving rotational freedom for the remaining fibres. As shown in Fig. 5, a step is formed on the bottom surface of each V-groove. A cylindrical spring element is placed above the MCF and is pressed downward by a pneumatic cylinder. When the pneumatic cylinder is actuated, the spring element undergoes elastic deformation and presses the MCF against the stepped portion of the V-groove, thereby fixing the fibre in position. The elastic deformation of the spring element allows

it to conform to slight diameter variations among individual MCFs. As a result, uniform fixation force can be applied to each fibre without requiring precise adjustment for each MCF. This structure enables individual fixation of each MCF at its optimum rotational position.

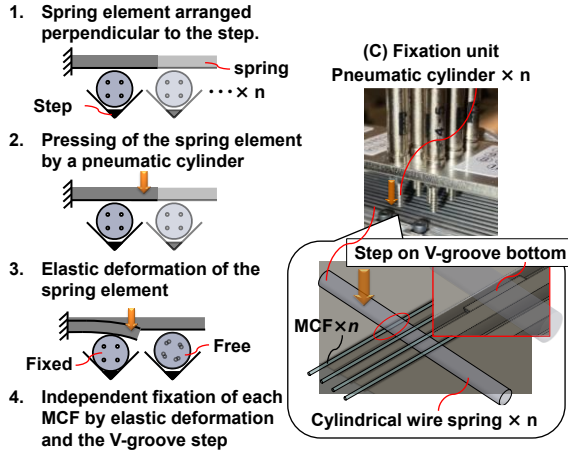


Fig. 5: Detailed schematic of the individual MCF fixation mechanism, where a step on the bottom surface of the V-groove and elastic deformation of a cylindrical spring element pressed by a pneumatic cylinder enable independent fixation of each MCF.

Importantly, the fixation operation for a specific MCF does not mechanically interfere with the rotational motion of the remaining unfixed fibres. This independent fixation mechanism is essential for realizing the simultaneous rotational alignment process described in Fig. 2(b).

Experimental result

The proposed simultaneous rotational alignment method was experimentally evaluated using a 12-fibre ribbon composed of 4-core MCFs, as shown in Fig. 3. Rotational alignment was performed while selectively fixing individual MCFs using the custom-fabricated alignment jig shown in Fig. 4.

Figure 6 shows the measured rotational misalignment of individual MCFs after completion of the simultaneous alignment process. The rotational misalignment was evaluated by comparing the angular position of each MCF at fixation with its optimum core-aligned orientation. The measured rotational misalignment exhibited an average value of 0.83° , with a minimum of 0.01° and a maximum of 2.11° . These results indicate that the proposed method achieves stable rotational alignment accuracy across all MCFs in the ribbon. During the alignment process, individual MCFs were successfully fixed at their optimum rotational positions without interrupting the collective rotation of the remaining unfixed fibres. No mechanical interference between fixed and unfixed MCFs was observed, confirming the effectiveness of the independent fixation mechanism.

These experimental results demonstrate that the proposed method enables scalable rotational alignment of MCF ribbons while maintaining the rotational accuracy required for low-loss interconnection.

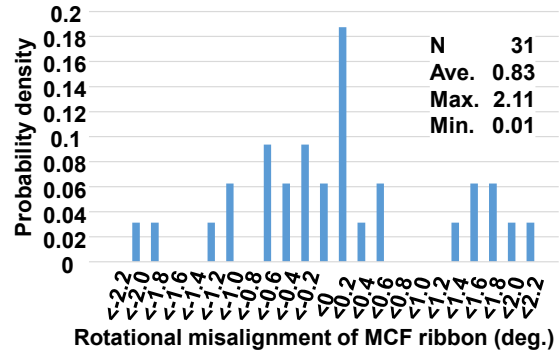


Fig. 6: Experimental result of simultaneous rotational alignment for MCF ribbon.

Conclusions

A simultaneous rotational alignment method for MCF ribbons has been proposed to enable scalable interconnection with reduced assembly time. The proposed method allows multiple MCFs to be rotated collectively, while individual fibres are selectively fixed at their optimum rotational positions using a dedicated alignment jig. Experimental results obtained using a 12-fibre ribbon composed of standard outer-diameter 4-core MCFs demonstrated an average rotational misalignment of 0.83° . These results indicate that the proposed method provides a meaningful level of rotational alignment accuracy while approaching the requirement for low-loss MCF interconnection. The independent fixation mechanism successfully prevented mechanical interference between fixed and unfixed fibres during the alignment process.

These results indicate that the proposed method effectively decouples the alignment process from the number of fibres and provides a practical approach for scalable assembly of MCF-based connectors and splicing interfaces.

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